

RealityCheck

LLM Hallucination Detection and Correction Framework

1. Problem Statement

Large Language Models (LLMs) such as those developed by OpenAI, Google, and Meta have demonstrated remarkable capabilities in natural language understanding and generation. However, these models frequently generate hallucinated information, where the system produces confident but factually incorrect statements.

This issue becomes problematic in high-stakes domains such as healthcare, legal advisory, cybersecurity analysis, and academic research. Existing mitigation strategies such as Retrieval Augmented Generation (RAG) reduce hallucinations by grounding responses in external knowledge sources but do not reliably detect or correct hallucinated claims within generated text.

The core research problem addressed in this work is:

How can hallucinated content in Large Language Model outputs be automatically detected and corrected while preserving response coherence and factual accuracy?

The objective of this research is to design a Hybrid Hallucination Detection and Correction Framework that integrates fact verification, confidence scoring, and retrieval-based correction to improve the factual reliability of LLM responses.

2. Relevance to Current Scenario

The rapid adoption of LLM-based systems across industries has created a critical challenge regarding trustworthiness and reliability of AI-generated information.

Applications currently affected include:

- medical diagnostic assistants
- financial advisory systems
- academic writing assistants
- automated customer support agents

In these scenarios, hallucinated responses can result in:

- misinformation
- incorrect decisions
- reduced trust in AI systems

Recent studies have shown that even advanced models may hallucinate 15–30% of factual responses in certain tasks. As AI systems are increasingly integrated into decision-making pipelines, detecting and correcting hallucinated outputs has become one of the most pressing challenges in AI safety and reliability research.

3. Expected Impact

The proposed framework aims to enhance the reliability of LLM-based systems by introducing an automated mechanism for hallucination detection and correction.

Potential impacts include:

- improved trustworthiness of AI assistants
- safer deployment of LLMs in healthcare and legal domains
- reduction of misinformation in AI-generated content
- improved robustness of AI-driven decision support systems

Additionally, the framework can be integrated into existing LLM pipelines as a post-processing verification layer, allowing organizations to deploy LLMs more confidently.

4. Dataset Availability

One of the advantages of this research problem is the availability of well-established benchmark datasets used in hallucination detection and factual verification research.

Potential datasets include:

1. TruthfulQA
2. A benchmark dataset designed to evaluate whether language models generate truthful answers to factual questions.

3. FEVER (Fact Extraction and Verification)

4. A large-scale dataset consisting of claims extracted from Wikipedia along with labels indicating whether the claims are supported, refuted, or unverifiable.

5. HaluEval Dataset

6. A specialized dataset specifically created to evaluate hallucinations in LLM-generated responses.

7. WikiBio Hallucination Dataset

8. Contains biography-related text where hallucinated statements can be detected against known facts.

These datasets provide a standard benchmark for comparing proposed methods with existing hallucination detection approaches.

5. Performance Metrics

To evaluate the effectiveness of the proposed hallucination detection and correction framework, the following metrics will be used.

Hallucination Detection Metrics

Precision : Percentage of detected hallucinations that are actually hallucinations.

Recall : Percentage of actual hallucinations successfully detected by the system.

F1 Score : Harmonic mean of precision and recall used to measure overall detection performance.

Correction Quality Metrics

Correction Accuracy : Percentage of hallucinated statements successfully corrected by the system.

BLEU / ROUGE Score : Used to evaluate similarity between corrected responses and ground truth factual responses.

Reliability Metrics

Fact Consistency Score : Measures the factual alignment of generated responses with verified knowledge sources.